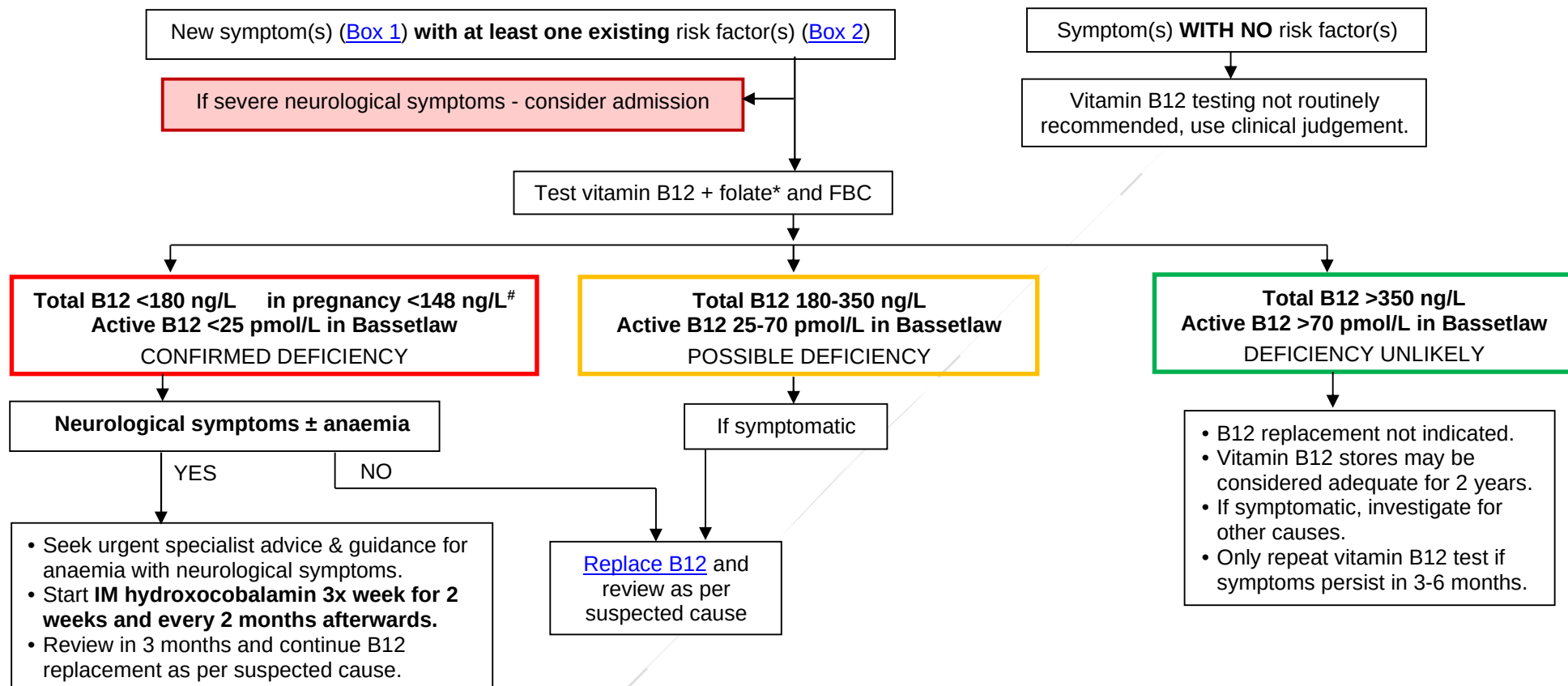


Vitamin B12 management in adults

NOTTINGHAMSHIRE GUIDELINES ON THE DIAGNOSIS AND MANAGEMENT OF VITAMIN B12 REPLACEMENT IN PRIMARY CARE

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SUSPECTED VITAMIN B12 DEFICIENCY



*In combined B12 and folate deficiency, always start B12 24-hours ahead of folate to protect neurones. In severe B12 or folate deficiency replacement may cause severe hypokalaemia as red cell production restarts.

#For pregnant patients, indicate pregnancy when requesting the test. Total B12 level falls physiologically during pregnancy. Seek urgent advice if neurological or haematological symptoms. If total B12 level >148ng/L and no neurological/haematological symptoms, B12 replacement is not generally required. IM hydroxocobalamin loading dose followed by the maintenance is the preferred treatment option in pregnancy. Symptom review is recommended in 1 month.

FURTHER INVESTIGATIONS WITH CONFIRMED B12 DEFICIENCY:

- Request **anti-intrinsic factor antibody** and **serological tests for coeliac disease** (if not tested positive before and no history of previous gastric operations).
- If autoimmune gastritis is suspected, despite negative anti-intrinsic factor antibody test, consider referral for further investigations.

CAUSE OF B12 DEFICIENCY	REPLACEMENT RECOMMENDATION with dietary advice to increase intake of foods rich in vitamin B12	DURATION	MONITORING
MALABSORPTION secondary to: • suspected or diagnosed autoimmune gastritis or pernicious anaemia, • total gastrectomy or complete terminal ileal resection.	IM hydroxocobalamin promote self-administration where possible LOADING: 1mg 3x week for 2 weeks MAINTENANCE: 1mg every 2 to 3 months Oral route is not recommended.	Lifelong	No routine B12 levels monitoring required unless presenting with new symptoms of deficiency.
OTHER GASTRIC CAUSES OF MALABSORPTION: e.g. coeliac disease, partial gastrectomy or terminal ileal resection, Roux-en-Y gastric bypass.	IM hydroxocobalamin promote self-administration where possible LOADING: 1mg 3x week for 2 weeks MAINTENANCE: 1mg every 2 to 3 months Oral cyanocobalamin 1mg daily only if IM is clinically inappropriate.		
CO-PRESCRIBED MEDICATIONS see list on page 4 OR RECREATIONAL USE OF NITROUS OXIDE (NO)	For patients who can self-administer: IM hydroxocobalamin LOADING: 1mg 3x week for 2 weeks MAINTENANCE: 1mg every 3 months If unable to self-administer: oral cyanocobalamin 50micrograms one tablet daily (taken between meals). If symptoms do not improve after 3 months, increase dose every 3 months in 50microgram steps up to 150microgram daily, or change to IM hydroxocobalamin 1mg every 3 months (administered in clinic).	Continue B12 replacement for as long as the causative medication is co-prescribed.	No routine B12 levels monitoring required unless presenting with new symptoms of deficiency. Once causative medication stopped, review B12 level and symptoms after 3 months.
DIETARY OR UNCERTAIN (when malabsorption not confirmed by investigations)	For dietary deficiency advise over-the-counter oral supplementation containing any of the following type of vitamin B12: cyanocobalamin, methylcobalamin, adenosylcobalamin. IM hydroxocobalamin promote self-administration where possible LOADING: 1mg 3x week for 2 weeks MAINTENANCE: 1mg every 3 months Only for patients with: • coexisting neurological or haematological condition which may deteriorate rapidly (e.g. ataxia, anaemia), • adherence issues (e.g. frail and in hospital, delirium, cognitive impairment, homelessness).	Review B12 level and symptoms after 3 months. If improved, replacement may be stopped with advice on ongoing self-care supplementation and diet.	

SYMPTOMS AND RISK FACTORS - WHEN TO TEST?

Offer vitamin B12 test for people with **at least one common symptom and at least one common risk factor** for deficiency.

Vitamin B12 testing is not routinely recommended for symptomatic people with none of the common risk factors, use clinical judgement.

Box 1

SYMPTOMS

are non-specific and may vary between individuals

- **Haematological**:
 - anaemia,
 - macrocytosis.
- **Neuro-psychiatric**:
 - **incoordination, balance problems, falls, impaired gait, pins and needles, numbness,**
 - **optic neuritis** e.g. blurred vision, optic atrophy, visual field loss,
 - **cognitive change** e.g. short-term memory loss, dementia, difficulty concentrating.
- **Oral**: glossitis (sore beefy red tongue) or angular cheilitis.
- Unexplained **fatigue**.

Box 2

RISK FACTORS

normal B12 absorption depends on sufficient dietary intake AND physiological processes in stomach and terminal ileum which may be affected by the following:

- Vitamin B12 **malabsorption** secondary to:
 - **atrophic gastritis** affecting the gastric body including **pernicious anaemia**,
 - **coeliac disease** or another **autoimmune condition** (e.g. thyroid disease, Sjögren's syndrome, inflammatory bowel disease, pancreatic failure, type 1 diabetes),
 - past gastro-intestinal surgery (e.g. **gastrectomy, terminal ileal resection**),
 - past abdominal or pelvic **radiotherapy**,
 - chronic **alcoholism**,
 - recreational use of **nitrous oxide**,
 - **family history** of B12 deficiency or an autoimmune condition.
- Co-prescribed **medications**:
 - colchicine,
 - H2-receptor antagonists,
 - metformin ([see the MHRA safety advice](#)),
 - phenobarbital,
 - pregabalin,
 - primidone,
 - proton pump inhibitors,
 - topiramate.
- **Insufficient dietary intake of vitamin B12**: (with no routine/regular supplementation of over-the-counter preparations containing vitamin B12) e.g.
 - restricted or vegan, vegetarian diets
 - limited or poor food intake (e.g. dementia, frailty, eating disorder or low income).

KEY NOTES

- On initiation of **IM B12** replacement, offer **self-administration training** if appropriate.
- On treatment initiation patients should be advised that **dose, formulation, and administration route** of vitamin B12 replacement **may be changed following clinical investigations**. This includes change of prescription only treatment (injectable or oral) to a recommendation for self-care supplementation with over-the-counter products.
- Advise patients that **symptoms may improve within 2 weeks up to 3 months** and may initially get worse but should improve with time. The response to treatment is variable and dependent on underlying cause.
- When investigating vitamin B level **in patients who use nitrous oxide (NO)**, NICE recommends serum methylmalonic acid (MMA) as the initial test. This is not available in Primary Care, please indicate use of NO within the test request if possible.
- **Interpret results with caution** in patients who take:
 - **over-the-counter vitamin B12 supplementation** - may mask untreated deficiency, if symptoms and low B12 level persist despite oral treatment, switch to IM B12 replacement.
 - **combined oral contraceptive pills** – can lower total vitamin B12 even in the absence of deficiency (altered binding to plasma protein). This should be considered for asymptomatic patients with incidental finding of possible B12 deficiency.
- Patients with known **cobalt allergy** should be advised to monitor for sensitivity reactions to cobalt contained in vitamin B12 and seek advice in case of allergic skin reactions e.g. rash or hives. Urgent medical attention is advised with symptoms of a serious allergic reaction e.g. extensive or blistering rash, wheeze, difficulty breathing, feeling faint. The hypersensitivity reaction may be immediate or delayed to 12-72 hours following vitamin B12 administration (oral and IM). There is no alternative treatment for vitamin B12 deficiency, therefore, vitamin B12 use is not contraindicated in patients with cobalt allergy that presents only as cutaneous symptoms. However, where previous serious allergic reaction is established in known cobalt allergy patients, individual assessment of the benefits and risks should be conducted before starting treatment. [See MHRA advice for more details.](#)
- When prescribing hydroxocobalamin IM injections for patients who **self-administer** or are to have it administered **at home by the community nursing team**, prescribers are reminded to **untick the 'personal administration' box on Systm One**. Unticking this box allows the prescription to be sent electronically to the pharmacy for dispensing and delivery/collection to/by the patient. For patients who are having the injection **administered within the GP practice**, leave the 'personal administration' box ticked. This will result in a paper FP10 being printed, which the practice can then use to claim back for the injection and administration.

WHEN TO REFER:

- Where IM injections are indicated but not tolerated or not possible - seek haematology advice.
- If there is a failure to respond to therapy i.e. there is not a prompt rise in Hb level. (Note: vitamin B12 levels do not require re-checking if the patient is already on parenteral replacement therapy. The FBC can be used to monitor response) - seek haematology advice.
- Consider referral to/discussion with appropriate specialty if vitamin B12/folate deficiency plus gastrointestinal disease or neurological symptoms or dementia.

SELF-ADMINISTRATION OF IM B12

There is no recommendation made within NICE guidelines, however self-administration of IM B12 may support patients care in the following ways:

- greater convenience with no administration appointments required (saves patients time and cost),
- promotion of self-care and autonomy (empowering patients to improve their own health),
- greater satisfaction (addressing patient preferences via shared decision making),
- reduced primary care burden allowing to re-direct resources and improve access to local healthcare system.

Self-administration may not be suitable for patients with reduced manual dexterity and limited physical or mental capacity, who do not have carers, representatives or family members who would be able to help.

To support self-administration of IM B12 replacement, patients will require the following (in addition to standard prescription for intra-muscular hydroxocobalamin):

Sharps bin	SharpsGuard Yellow 1	prescribed on FP10
2ml syringe for injection	i.e. Emerald BD	provided from the practice stock
Injecting needle	Gauge: 22-25, length: 1-1.5 inches i.e. blue needle (or green if longer needle required)	
Needle for drawing up	i.e. purple needle	

Patients should be instructed and assessed for the self-administration by a practice nurse on the first B12 injection appointment at the treatment initiation.

The following resources are available from various NHS organisations to further support patients:

- [B12 Injections - Eastwood Primary Care Centre](#)
- [How to self-inject intramuscular vitamin B12 - Overview | Guy's and St Thomas' NHS Foundation Trust](#)

PATIENT ADVICE

Foods containing vitamin B12

The following foods are good sources of vitamin B12. Including these foods regularly in the diet should help to prevent vitamin B12 deficiency:

- liver/liver pâté (note that pregnant women should avoid liver/liver pâté),
- eggs,
- cheese,
- milk,
- meat - for example, beef, lamb, and pork,
- fish,
- fortified breakfast cereals,
- Marmite[®],
- fortified alternatives to dairy milk e.g., goat, rice, and soya milks,
- fortified soya yoghurts,
- fortified spreads,
- fortified yeast extract.

People who are vegan should aim to include foods that are fortified with vitamin B12, at least three times a day. If these foods are not consumed in adequate amounts, the Vegan Society recommends a vitamin B12 supplement which can be purchased from a pharmacy or health food shop.

PATIENT INFORMATION RESOURCES:

- General information on vitamin B12 and deficiency: <https://www.nhs.uk/conditions/vitamins-and-minerals/vitamin-b12/>
- Advice on taking oral cyanocobalamin: <https://www.nhs.uk/medicines/cyanocobalamin/how-and-when-to-take-cyanocobalamin/>
- Treatment information for patients prescribed oral vitamin B12: <https://www.nhs.uk/medicines/cyanocobalamin/how-and-when-to-take-cyanocobalamin/>
- Treatment information for patients prescribed IM vitamin B12: <https://www.nhs.uk/medicines/hydroxocobalamin/>

- Information and advice on vitamin B12 for vegan patients: <https://www.vegansociety.com/resources/nutrition-and-health/nutrients/vitamin-b12>
- Patient information on pernicious anaemia and vitamin B12: <https://patient.info/allergies-blood-immune/anaemia-leaflet/vitamin-b12-deficiency-and-pernicious-anaemia>

REFERENCES AND FURTHER RESOURCES FOR CLINICIANS:

- [Overview | Vitamin B12 deficiency in over 16s: diagnosis and management | Guidance | NICE](#)
- [Vitamin B12 \(hydroxocobalamin, cyanocobalamin\): advise patients with known cobalt allergy to be vigilant for sensitivity reactions - GOV.UK \(www.gov.uk\)](#)
- [Metformin and reduced vitamin B12 levels: new advice for monitoring patients at risk - GOV.UK \(www.gov.uk\)](#)
- [Vitamin B12 deficiency: treatment during pregnancy – SPS - Specialist Pharmacy Service – The first stop for professional medicines advice](#)
- [Scenario: Management | Management | Anaemia - B12 and folate deficiency | CKS | NICE](#)
- [Management of Obstetric Anaemia \(Secondary Care guideline\) – Family Health, Obstetrics and Maternity, Nottingham University Hospitals NHS Trust](#)